



Build a Virtual Private Cloud (VPC)



yakubu abdullahi yusuf

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

NextWork VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me



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Introducing Today's Project!

In this project, I will demonstrate. how to create a VPC. I'm doing this project to learn the networking concepts in cloud computing.

What is Amazon VPC?

Amazon VPC is a service that allows private network to be created in the cloud where organization's store their resources privately. and it is useful because Amazon VPC is useful because it gives you a private and secure place to run your cloud resources while still being able to connect to the internet when needed.

In today's project, I used Amazon VPC to create a secure network, set up subnets and an internet gateway, and allow my resources to communicate with the internet safely.

Personal reflection

This project took me 30 minutes to complete.



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One thing I didn't expect in this project was how the creation of these resources was faster using commands.



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Virtual Private Clouds (VPCs)

What I did in this step

In this step, I will access the AWS VPC console to create a VPC.

How VPCs work

VPCs are private network created in the cloud where organization's store their resources privately.

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This is because AWS automatically creates a default VPC in every region so users can quickly launch resources without having to manually set up networking.



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☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR provided by me

Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is 10.0.0.0/16



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Subnets

What I did in this step

In this step, I will create a subnet inside my VPC.

Creating and configuring subnets

Subnets are smaller sections of a VPC that help organize and control where resources are placed. There are already subnets existing in my account, one for every Availability Zone in the region.

Public vs private subnets

The difference between public and private subnets is that public subnets allow resources to access the internet directly, while private subnets do not. For a subnet to be considered public, it has to be connected to an Internet Gateway.



You have successfully changed subnet settings:

◦ Enable auto-assign public IPv4 address

Subnets (1/4) [Info](#)

Last updated 17 minutes ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Publi
☐	-	subnet-0fd811de33df6e350	Available	vpc-003fffe87866a97ab	Off
☐	-	subnet-0438e58ff77c21010	Available	vpc-003fffe87866a97ab	Off
☐	-	subnet-0c63e359749539b68	Available	vpc-003fffe87866a97ab	Off
☒	Public 1 -subnet	subnet-0413f5192d99a28ec	Available	vpc-016d9b867bab64aff Next...	Off

Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled auto-assign public IPv4 addresses. This setting makes sure that any resource launched in the subnet automatically receives a public IP address, so that it can communicate directly with the internet.



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Internet gateways

What I did in this step

In this step, I will connect my VPC to the internet using internet gateway.

Setting up internet gateways

Internet gateways are like bridges that connects resources to the internet.

Attaching an internet gateway to a VPC means resources in my VPC can now access the internet, If I missed this step my resources would not be able to access the internet.



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Internet gateway igw-065c0f068bae50abf successfully attached to vpc-016d9b867bab64aff

igw-065c0f068bae50abf / NextWork IG

Actions

Details info

Internet gateway ID igw-065c0f068bae50abf	State Attached	VPC ID vpc-016d9b867bab64aff NextWork VPC	Owner 176971016201
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Tags (1)

Search tags

Key	Value
Name	NextWork IG

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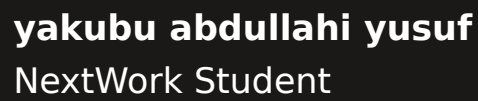
Using the AWS CLI

What I'm doing in this extension

In this project extension, I will run AWS commands in the CloudShell to set up VPC, subnets and internet gateways.

Debugging my setup

To set up a VPC or a subnet, you can use the command `aws ec2 create-vpc --cidr-block 10.0.0.0/24 --query Vpc.VpcId --output text` or ```bash aws ec2 create-subnet --vpc-id --cidr-block --availability-zone`, Make sure to avoid errors by including `--cidr-block` and `--availability-zone`.

[illegible]

Comparing CloudShell vs AWS Console

Compared to using the AWS Console, an advantage of using commands is they are faster. An advantage of using the Console is that it is visual and easier to understand, especially for beginners. Overall, I preferred using commands.



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